「Summary

$$\text{Maximal} \Leftrightarrow \text{Prime} \Leftrightarrow \text{Irreducible}.$$

Bézout
$\cancel{\Leftarrow}$
Bezout
I.D
I.D

$x \in \mathbb{Z}[x,y]$
$2 \in \mathbb{Z}[x]$
$2 \in \mathbb{Z}[\sqrt{-5}]$

$$\text{Maximal} \Rightarrow \text{Prime} \Rightarrow \text{Irreducible}.$$

Noether domain $\Rightarrow$ Atomic domain.

If irr = prime.

Bézout domain

$\Uparrow$

Field $\Rightarrow$ Euclidean Domain $\Rightarrow$ Principal Ideal Domain $\Rightarrow$ Unique Factorization Domain $\subset$ Integral Domain.

P.I.D
P.I.D.

$$\text{Maximal} \Leftrightarrow \text{Prime} \Leftrightarrow \text{Irreducible}.$$

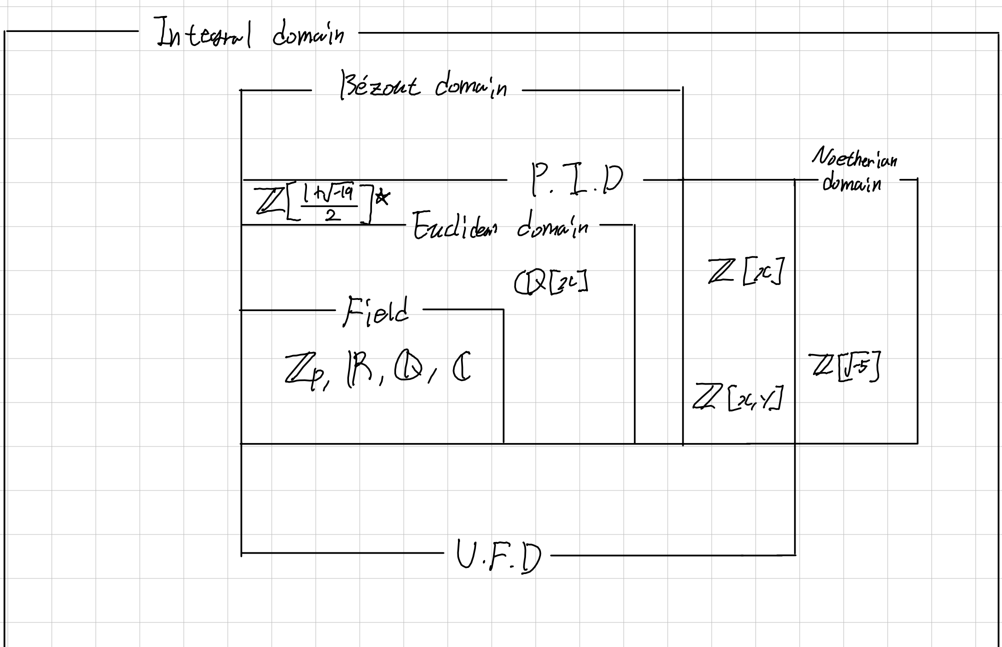
I.D
I.D

U.F.D
$2 \in \mathbb{Z}[x]$
U.F.D.

$$\text{Maximal} \Rightarrow \text{Prime} \Leftrightarrow \text{Irreducible}.$$

P.I.D → Noether $\Leftrightarrow$ Every Ideal are finitely generated $\Leftrightarrow$ A.C.C.I $\Rightarrow$ A.C.C.P $\Rightarrow$ $\exists$ factorization → U.F.D

P.I.D → Bézout $\Leftrightarrow$ Finitely generated ideal is principal $\Rightarrow$ irreducible = Prime $\Rightarrow$! factorization → U.F.D



「Basic languages for generated ideal.

「Def. Let R be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1. a is called a multiple of b if: there exists $x \in R$ s.t $a = bx$. Simply, $b | a$.
2. A non-zero $d \in R$ is called a greatest common divisor of a and b if;

$$d|a \text{ and } d|b \ (\text{or, } a, b \in (d)), \quad d'|a \text{ and } d'|b \Rightarrow d'|d.$$
$$(a, b \in (d') \Rightarrow (d) \subseteq (d'))$$

Note: $b|a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$.

Directly, $(a,b) = (d)$ implies d is the g.c.d of a and b,

but Conversely, gcd maybe does not generate the ideal; in $\mathbb{Z}[x]$,

$$(1) \neq (2, x) \quad \text{but } 1 \text{ is gcd of } 2 \text{ and } x$$

「Prop. Let R be a Commutative Ring, and $a, b, c \in R$ be given. Then,

1). $(a, b) = (a) + (b)$

2). $(c) \cdot [(a) + (b)] = (c) \cdot (a) + (c) \cdot (b) = (ca) + (cb)$.

3).